

Efficient, Reliable and Secure Framework for the 5G Virtualization of the UltraSound Medical System

Alessandro Carrega* Roberto Bruschi* Raffaele Bolla* Antonio Passalacqua[§]

* Department of Electrical, Electronic and Telecommunications Eng., and Naval Architecture (DITEN)
University of Genoa (UniGe), Italy {name}. {surname}@unige.it

[§] ESOATE, Genoa, Italy {name}. {surname}@esaote.com

Abstract—The virtualization of UltraSound (US) medical imaging systems through cloud-edge computing offers a transformative approach to healthcare, overcoming limitations of traditional on-premises systems. By dematerializing US system functions and migrating them to the cloud, barriers related to hardware and localization are eliminated, enabling a more agile and scalable approach to medical imaging. This paradigm shift leverages virtual objects and composite virtual objects to intelligently manage and orchestrate physical and virtual components, optimizing system performance and meeting service requirements.

Index Terms—Framework, Virtualization, UltraSound Medical System, 5G, NFV, Security

I. INTRODUCTION

The current UltraSound (US) medical imaging processes are constrained by both the technical features of the local device and the knowledge of the (local) healthcare operator performing the examination. In fact, Electronic Health Record (EHR) processes are currently bound to on-premises dedicated hardware/firmware components to fulfil the need of a real-time or an almost real-time execution of the process [1].

As a result, acquisition costs/capital expenses are very high and limit the degrees of flexibility in upgrading the hardware and, consequently, the types and number of functions that can be (locally) provided. [2] Functions refer to those EHR processes that elaborate US data to provide the operator with additional qualitative information or quantitative data.

The goal of this Use Case (UC) is to connect, and somehow to decompose and virtualize US medical imaging systems into the cloud-edge continuum to lose any barriers due to the hardware capabilities and localization of physical systems.

By exploiting and leveraging on Fifth-Generation (5G) and Internet of Things (IoT) technologies, the idea is to transform the US acquisition hardware and the medical imaging viewers into smart wireless-connected “things”, that can be “plugged and played” through the cloud-edge medical imaging application: the essential functions of the US system, with the sole exception of the probe and the input/output devices (such as monitors, keyboards, etc.), must be dematerialized and migrated to the cloud/edge.

The ultrasound image processing currently involves a probe, image acquisition hardware, several local software functions devoted to the actual image processing and a monitor for displaying the images. The probe is a passive element cabled to the acquisition hardware in the device. The same device locally performs base and advanced image processing (typically using embodied Graphical Processing Units (gpus)) and renders the results on the local monitor. Since the image should have a high medical-grade resolution and that the whole imaging process is very complex, as it requires high volumes

of data to be processed with strict latency (to react to human/operator-driven actions) and security constraints, present-day systems heavily rely on HardWare (hW)/FirmWare (FW) tools. The decomposition of such systems into the cloud-edge continuum encompasses: (a) the management through the Virtual Object (VO) stack of the connected physical acquisition and rendering devices, (b) the possibility to “plug and play” physical systems (by means of their VOs) into different instances of the ultrasound medical app to execute visits even by involving remote operators (with their monitors and keyboards), (c) the possibility of smartly manage (as-a-Service and at runtime) the processes for added value qualitative/quantitative analysis, medical reporting, and hardware maintenance [3].

The virtualized components, along with additional processes, will be deployed across the edge-cloud continuum, depending on the strictness of their time requirements. In more details, the base ultrasound image acquisition and visualization will become likely a tactile Internet application, and as such its proximity to the physical acquisition/rendering system will be crucial to provide the needed reactivity to the operator actions. On the other hand, the processes related to the medical reports’ generation are less time-critical and can be deployed in the cloud. Between these two categories of applications, the edge-cloud continuum is accomplished with several services which still have strict latency requirements, but not as in the tactile Internet realm. The most relevant of these applications regard the overlay processes in charge of elaborating the raw images to identify known patterns or perform measurements, which are heavily based on Machine and Deep Learning - Machine Learning (ML) and Deep Learning (DL) - techniques [4].

Finally, the current system capabilities will be improved by pairing the physical components and, potentially, some of the virtualized functions, with a VO. Such digital counterparts will support and extend the capabilities of the IoT devices as well as helping with the interplay of physical and virtualized processes, for example, by adapting the image coding to the monitor resolution, providing data pre-processing, by managing caching, etc. [5].

A. Benefits

There are several possible advantages considering the goal described in the previous section. In particular, the benefits can be grouped in different categories based on the possible beneficiaries. From the point of view of the clinical device manufacture (e.g., ESAOTE) there is a dramatic cost reduction: no plastics, mechanical boards, spare parts. The focus is on the software transducers and high parallel computing network. There is also a dramatic reduction of transport and installation cost and service management and maintenance. An additional benefit could be a less environmental impact.

Instead, from the point of view of the clinical staff and the hospital there is the possibility to use always up-to-date equipment with the support of remote control and diagnosis. This solution also allows a space reorganization.

Finally, there are benefits regarding the reduced time for reporting and training for the clinical staff and the end-user (in this specific case: the patients).

II. TECHNICAL REQUIREMENTS AND CHALLENGES

The distribution of ultrasound medical system into different application components in the cloud-edge continuum poses several challenges related heterogeneous performance levels required by the different functions (falling from “*Tactile Internet*” requirements to ones generally provided by current cloud systems), and to the way data is treated. Data security is of paramount importance for medical processes and so is the need of a real-time or an almost real-time execution of the process.

To best face the needs of the EHR operations in the use case above and offer solutions to reach the overall goal for the solution we can identify the following main technical requirements and challenges.

Orchestration of software components: given the EHR application graph a dynamic placement of software components should be enabled based on service requirements and resource availability.

Device Management: some application functionalities can be pre-deployed on the devices or at the edge.

Low latency communication: communication networks to/from a rural environment towards the edge and cloud should guarantee low delays for fast response under mobility conditions and possible disconnections.

High bandwidth for edge/cloud: the data collected from the probe and after some pre-processing with the Minimal HW Device (mHWDDev) should be sent to the edge/cloud for advanced analysis and to obtain additional diagnosis from remote and skilled operators.

Smart data filtering/aggregation/compression: large amount of data is collected from the probe and it can be filtered, other ones can be down sampled or aggregated before sending it to the edge/cloud using the mHWDDev for Local Processing (LP).

III. APPLICATIONS AND SERVICES OF THE STUDY CASE

A. Application Scenario: Real-Time Cloud Elaboration

This scenario refers to the application components and services to provide Real-Time (RT) elaboration of the data collected the different devices (probe and keyboard). The collected data using the probe and some command sent with the keyboard are sent to the cloud for additional elaboration. The mHWDDev is responsible to make preliminary elaboration.

For different VOs should be deployed for the following IoT devices: Primary Screen (PS), mHWDDev, Touchscreen Display (TD), and the GateWay (GW) that is used to send the command from the Keyboard to the cloud. The Keyboard and the Probe devices are not directly connected to the network, and, for this reason, a specific VO is not required.

A network connection fulfilling data rate and latency requirements for data streaming is required between the mHWDDev and A Lightweight Software Stack And Synergetic Meta-Orchestration Framework For The Next Generation Compute Continuum (NEPHELE) through the corresponding VO to send the data and process them. The communication between physical devices, the virtual counterparts at the VO and the other application components

is enabled through the *Zenoh* [6]. protocol (some data communication can be integrated using HyperText Transfer Protocol (HTTP) and Representational State Transfer (REST) implementation). Some service will be running on the physical devices, whereas other on the edge and cloud continuum and will have to be configured through the VO.

The mHWDDev is responsible to decrease the amount of data sent to the cloud for the processing. In addition, it includes local storage to save the data in case the connection is lost and should be sent when the connection is recovered.

Figure 1 shows the application graph for the RT Cloud Elaboration application scenario.

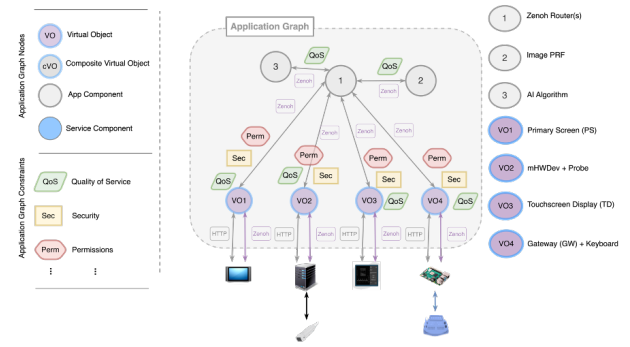


Figure 1. RT Cloud Elaboration application scenario - Application graph.

B. Application Scenario: Remote Support

This scenario refers to the application components and services to provide remote support for maintenance, tutorial, and training activities. The elaborated data is accessible in RT using a dashboard on a Web Interface. Like the previous application scenario, a VO should be deployed the following IoT devices: PS, mHWDDev, TD, and the GW.

A network connection fulfilling data rate and latency requirements for data streaming is required between the mHWDDev and NEPHELE through the corresponding VO to send the data and process them. The communication between physical devices, the virtual counterparts at the VO and the other application components is enabled through the *Zenoh* protocol. Some service will be running on the physical devices, whereas other on the edge and cloud continuum and will have to be configured through the VO.

The Web interface includes a Dashboard that allows the following remote operations: monitoring, alerting, and replaying. In addition, with the Dashboard is it possible to make further remote elaboration for advanced analysis.

Figure 2 shows the application graph for the Remote Support application scenario.

C. Application Scenario: Off-Line Remote Consultation

This scenario refers to the application components and services to provide a data storage needed to perform off-line consultation of the elaborated data using a Web Interface. The web interface makes possible to program further elaboration that can be useful for maintained, tutorial and training activities in a similar way of the previous application scenario.

A network connection fulfilling data rate and latency requirements for data streaming is required between the mHWDDev and NEPHELE through the corresponding VO to send the data and process them. The communication between physical devices, the virtual counterparts at

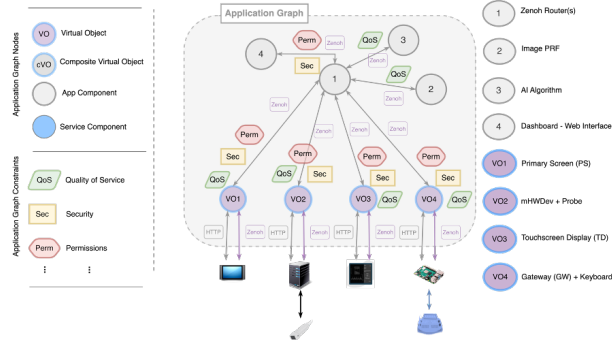


Figure 2. Remote Support application scenario - Application graph.

the VO and the other application components is enabled through the *Zenoh* protocol. Some service will be running on the physical devices, whereas other on the edge and cloud continuum and will have to be configured through the VO.

The Data storage includes a Time Series DataBase (DB) that allows to view the history of the elaborated data that can be used to simulate a stored cases useful to make additional analysis and elaboration.

Figure 3 shows the the application graph for the Off-Line Remote Consultation application scenario.

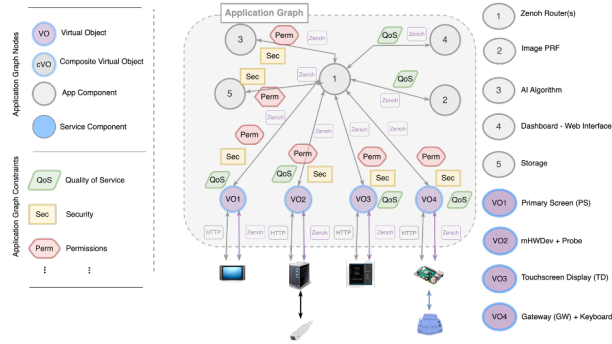


Figure 3. Off-Line Remote Consultation application scenario - Application graph.

IV. APPLICATION COMPONENTS

The EHR Hyper Distributed Application (HDA) will have a classic three-tier architecture with a *presentation* tier, an *application* tier, and a *data* tier.

Presentation tier: the application will offer a frontend for visualization of the clinical situation of the patient. The dashboard integrates data coming from the probe together related to the intervention area and the patients' personal information. The dashboard will be accessible through a web browser or a Graphical User Interface (GUI) remotely and enable the Service Consumer to interact with the *application* tier to see the clinical status of the patient and analyse historical data.

Application tier: the inputs and requests coming from the *presentation* tier are collected and application components are activated to execute intervention operations. At this level, all application components for supporting the application logic in this UC are included. Some of these components will run directly on the IoT devices, some on the edge and some on the cloud through the VO/Composite VO (cVO).

Data tier includes a storage element for storing processed images or historical data about the EHR intervention. The data produced by the probes will be compressed, down sampled and/or secured before being stored for future use by the *application* tier.

V. CONCLUSIONS

The paper provides a comprehensive overview of the integration of 5G and IoT technologies in the healthcare sector, specifically focusing on the virtualization of US medical imaging systems. By leveraging cloud-edge computing capabilities, the traditional hardware components of US systems are transformed into smart, wireless-connected entities that can be seamlessly integrated into medical imaging applications. This transformation not only enhances the flexibility and functionality of US systems but also opens up new possibilities for remote diagnostics, real-time data analysis, and improved patient care.

In conclusion, the virtualization of US medical imaging systems through the integration of 5G and IoT technologies represents a significant advancement in the healthcare industry. By embracing cloud-edge computing, virtualization, and smart data processing techniques, healthcare providers can offer more personalized, efficient, and reliable diagnostic services to patients. The seamless integration of physical and virtual components, coupled with advanced data analysis capabilities, paves the way for a new era of healthcare delivery that is driven by innovation, connectivity, and intelligence.

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